

- **Vector and knowledge layer:** Vector databases and ontologies that power agentic AI and semantic search are foundational to genAI.
- **Feature and model stores:** Reusable features, a model registry, and model artifact storage enable machine learning models to be built, versioned, and reused consistently rather than recreated per project.
- **Content and document stores:** A repository that supports genAI applications operating directly over enterprise content (contracts, policies, knowledge articles).

AI, analytics and decision

intelligence: In this layer, the governed data is converted into insight, prediction, and increasingly autonomous action.

- **Descriptive and diagnostic:** Comprises BI (business intelligence), dashboards, and self-service analytics.
- **Predictive and prescriptive:** Includes machine learning models, optimisation, and simulation.
- **GenAI and agentic AI:** Copilots, task-oriented agents, and RAG pipelines generate content to take bounded actions on the enterprise's own data.
- **Decision intelligence:** Composite decision flows that blend rules engines, analytics, and AI models into a single decision path.

Business consumption and

experience: The components and agents in this layer inform the AI/analytics layer in real-time what gets built upstream.

The business layer brings together line-of-business applications, AI copilots and agents, automation, and performance tracking to drive outcomes. Business applications support daily operations and customer interactions, while embedded copilots and agents enhance productivity. Automation through BPM, RPA, and event-driven workflows reduces manual effort, and KPIs with outcome tracking ensure the solution delivers measurable business value.

Data management and semantics

layer: This layer makes data trustworthy, findable, and consistently defined. It is applied continuously across every stage of the pipeline rather than as a single processing step.

Governance, security and AI

TRISM: This layer ensures privacy, identity, and full traceability of data and models.

Top open source data security tools that are widely used include Metasploit, OSSEC, OpenVAS, Snort, KeePass, and ClamAV. These tools can be integrated into various security strategies to protect a wide range of cyber threats.

Platform engineering and MLOps/

DataOps: Modern data platforms combine DataOps, MLOps, platform engineering, and scalable infrastructure. DataOps automates testing and deployment of data pipelines, while MLOps manages model deployment, monitoring, and retraining. Platform engineering provides self-service tools and governance for faster, consistent deployments. Together with serverless computing, storage optimization, and cost management, these capabilities create efficient, scalable, and reliable data platforms.

Popular open source tools for data storage are Hadoop, LakeFS, Cassandra, and Neo4J. These provide scalability, robustness and performance in managing large

data and analysing large datasets in various applications.

Benefits of data governance

Enterprises with mature governance capabilities experience higher AI model accuracy, increased data trust, better decision-making, faster innovation cycles, improved compliance posture, reduced data duplication, and greater business agility.

Effective data governance ensures consistent, accurate, and trustworthy data across the enterprise, enabling better decision-making and stronger business outcomes. It establishes clear standards for data quality, governance, and workflow management while improving agility, scalability, and productivity through streamlined processes and data reuse. Centralised governance reduces duplication, lowers data management costs, optimises storage, minimises security risks, and builds confidence in data accuracy and documentation. It also helps organisations comply with regulatory requirements such as GDPR, CCPA, HIPAA, and PCI-DSS.

Today, data governance is not optional but mandatory. Effective data governance is a collection of processes, people, policies, standards, and metrics that ensure the effective and efficient use of data in enabling an enterprise to achieve its goals. **END** 

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